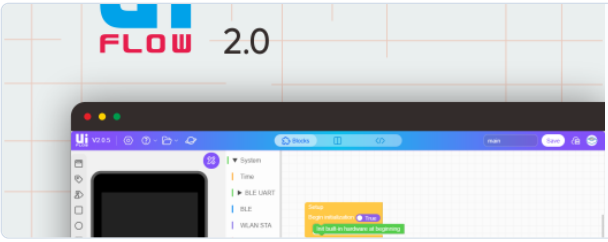
Arduino Nesso N1

SKU:DK001

Description

Arduino Nesso N1 is a high-performance all-in-one development board jointly developed by M5Stack and Arduino, designed for remote monitoring and automation control. The device uses ESP32-C6 SoC, supporting 2.4 GHz Wi-Fi 6, Thread, Zigbee, and Matter communication protocols. It also integrates an SX1262 LoRa transceiver, supporting 850 ~ 960 MHz frequency bands. Equipped with a 1.14" LCD touch screen, built-in 6-axis IMU sensor, programmable button, and rechargeable battery, it is an ideal tool for developing complex IoT solutions.

Tutorial



UiFlow2

This tutorial will introduce how to control the Arduino Nesso N1 device through the UiFlow2 graphical programming platform.

Arduino IDE

This tutorial will introduce how to program and control the Arduino Nesso N1 device using the Arduino IDE.

Features

- ESP32-C6 Core Controller:
  - 2.4 GHz Wi-Fi 6
  - Thread, Zigbee, Matter
- LoRa Communication:
  - SX1262 LoRa Transceiver
  - Operating Frequency 850 ~ 960 MHz
- Built-in 3.7V @ 250 mAh Battery
- 1.14" LCD Touch Screen
- 6-axis IMU Sensor BMI270
- Infrared Transmitter
- Buzzer
- Status Indicator Light
- Expansion Interface:
  - 2.54-8P Bus Interface (compatible with M5Stack Hat expansion series)
  - 1x Qwiic Interface

Includes

- 1 x Arduino Nesso N1
- 1 x Hex Key L-Shape 1.5mm (For M2 Screw)

Applications

- Remote Monitoring
- Industrial Automation
- IoT Application Development

Specifications

Specification	Parameter
SoC	ESP32-C6 @ RISC-V 32-bit high-performance single-core processor 160MHz + RISC-V 32-bit low-power single-core processor 20 MHz
Flash	16 MB
Wi-Fi	2.4 GHz Wi-Fi 6
Input Power	DC 5V
Battery	3.7V @ 250 mAh
LoRa	SX1262
Screen	ST7789 (1.14" IPS LCD) Resolution: 135 x 240
Screen Touch	FT6336
Operating Temperature	0 ~ 40°C

Learn

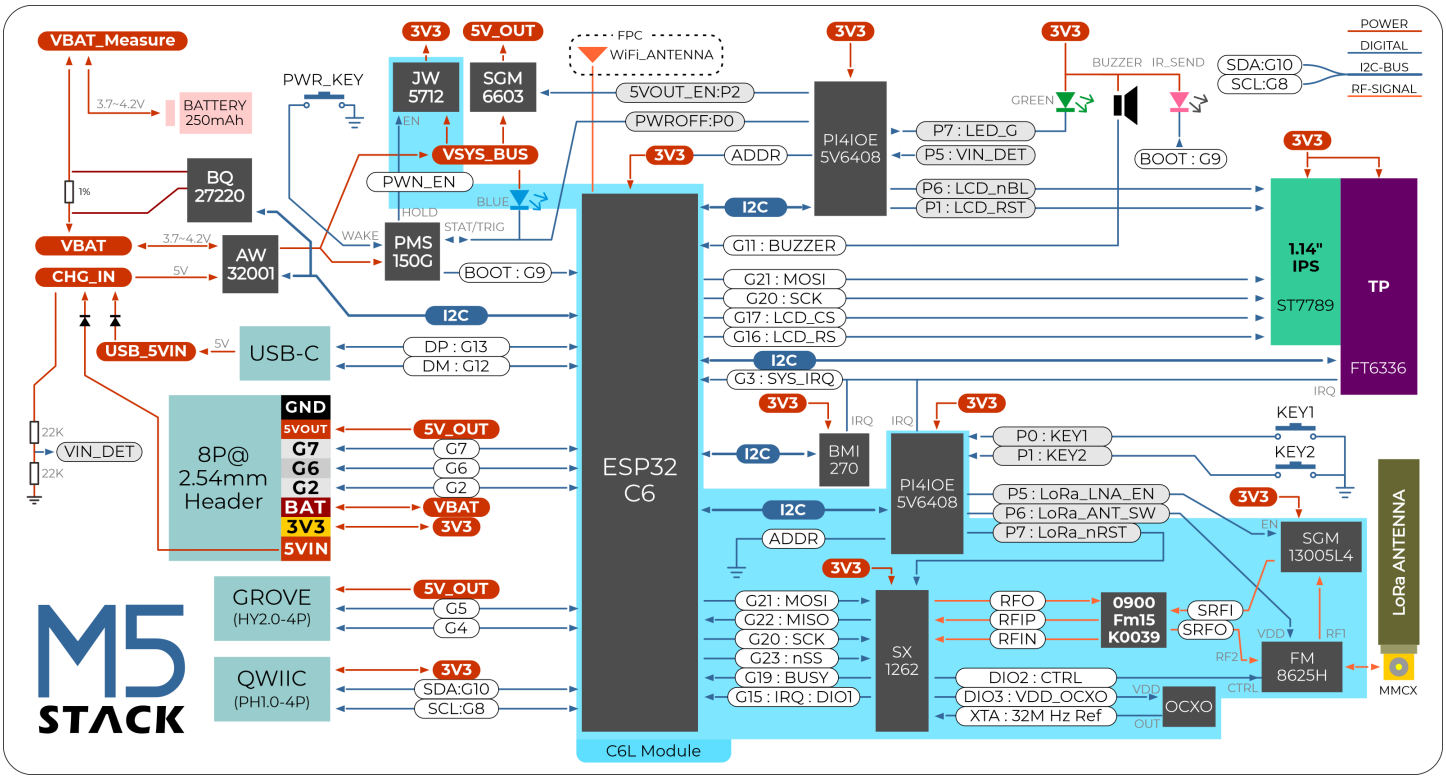
Button Operation

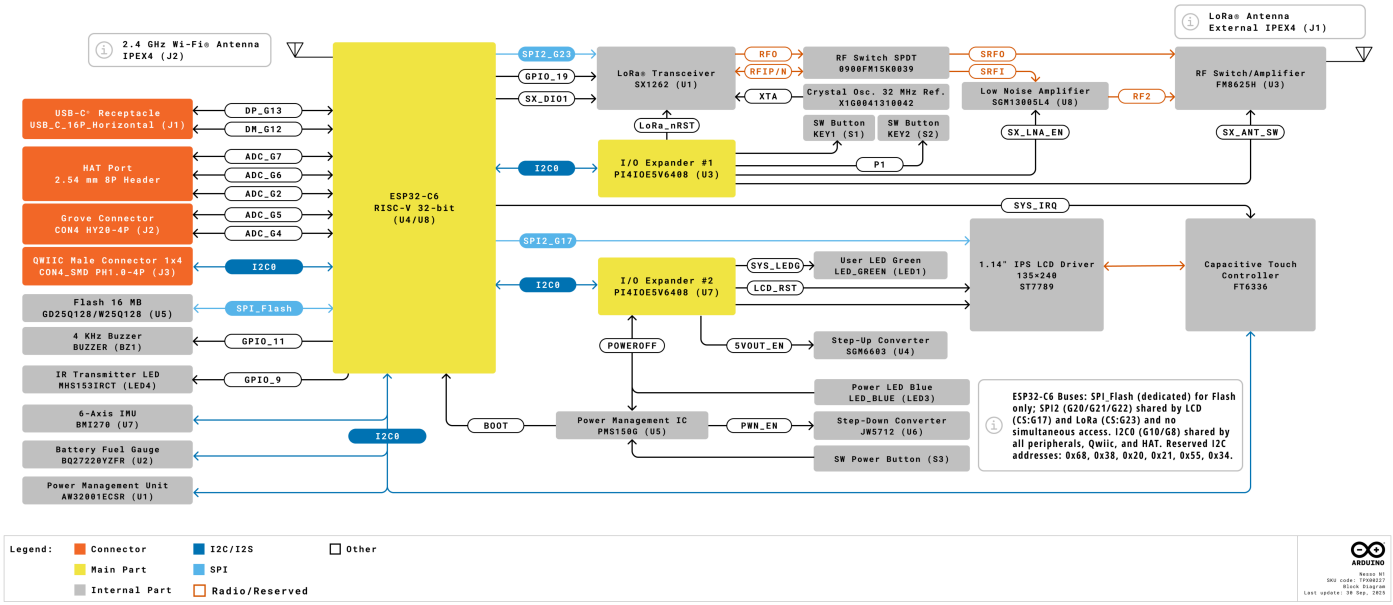
- Single Click: Power on or Reset the device
- Double Click: Power off the device
- Long Press: Enter Program Download Mode

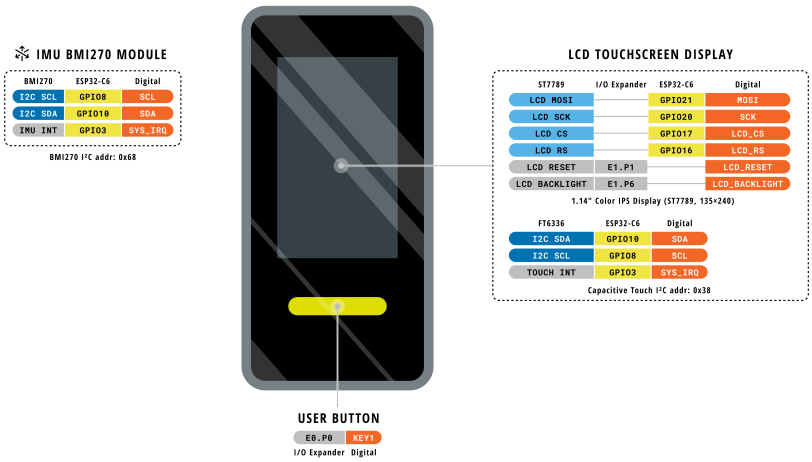
Download Mode

Connect the device with a USB cable and press and hold the reset button on the left side of the device body. When the internal blue LED starts flashing, it indicates that the device has successfully entered download mode.

Schematics


VBAT_Measure
3.7~4.2V
BATTERY 250mAh
PWR_KEY
3V3
5V_OUT
JW 5712
SGM 6603
EN
VSYS_BUS
FPC
WiFi_ANTENNA
5VOUT_EN:P2
PWROFF:P0
PWN_EN
HOLD
BLUE
WAKE
PMS 150G
STAT/TRIG
BOOT : G9
BQ 27220
1%
VBAT
3.7~4.2V
AW 32001
CHG_IN
5V
I2C
USB_5VIN
USB-C
DP : G13
DM : G12
22K
VIN_DET
8P@ 2.54mm Header
GND
5VOUT
G7
G6
G2
BAT
3V3
5VIN
ESP32 C6
C6L Module
ADDR
PI4IOE 5V6408
P7 : LED_G
P5 : VIN_DET
P6 : LCD_nBL
P1 : LCD_RST
BUZZER
IR_SEND
GREEN
SDA:G10
SCL:G8
POWER
DIGITAL
I2C-BUS
RF-SIGNAL
G11 : BUZZER
G21 : MOSI
G20 : SCK
G17 : LCD_CS
G16 : LCD_RS
G3 : SYS_IRQ
1.14" IPS
ST7789
TP
FT6336
IRQ
BMI 270
KEY1
KEY2
P0 : KEY1
P1 : KEY2
P5 : LoRa_LNA_EN
P6 : LoRa_ANT_SW
P7 : LoRa_nRST
SGM 13005L4
LoRa ANTENNA
MMCX
G21 : MOSI
G22 : MISO
G20 : SCK
G23 : nSS
G19 : BUSY
G15 : IRQ : DIO1
SX 1262
RFO
RFIP
RFIN
0900 Fm15 K0039
SRFI
SRFO
DIO2 : CTRL
DIO3 : VDD_OCXO
XTA : 32M Hz Ref
OCXO
FM 8625H
RF1
RF2
VDD
CTRL
OUT
GROVE (HY2.0-4P)
QWIIC (PH1.0-4P)
G5
G4
M5 STACK


2.4 GHz Wi-Fi® Antenna IPEX4 (J2)
LoRa® Antenna External IPEX4 (J1)
USB-C® Receptacle USB_C_16P_Horizontal (J1)
HAT Port 2.54 mm 8P Header
Grove Connector CON4 HY20-4P (J2)
QWIIC Male Connector 1x4 CON4_SMD PH1.0-4P (J3)
Flash 16 MB GD25Q128/W25Q128 (U5)
4 KHz Buzzer BUZZER (BZ1)
IR Transmitter LED MHS153IRCT (LED4)
6-Axis IMU BMI270 (U7)
Battery Fuel Gauge BQ27220YZFR (U2)
Power Management Unit AW32001ECSR (U1)
DP_G13
DM_G12
ADC_G7
ADC_G6
ADC_G2
ADC_G5
ADC_G4
I2C0
SPI_Flash
GPIO_11
GPIO_9
ESP32-C6 RISC-V 32-bit (U4/U8)
SPI2_G23
GPIO_19
SX_DIO1
LoRa® Transceiver SX1262 (U1)
LoRa_nRST
RFO
RFIP/N
XTA
RF Switch SPDT 0900FM15K0039
Crystal Osc. 32 MHz Ref. X1G0041310042
SRFO
SRFI
Low Noise Amplifier SGM13005L4 (U8)
RF2
RF Switch/Amplifier FM8625H (U3)
SW Button KEY1 (S1)
SW Button KEY2 (S2)
P1
SX_LNA_EN
SX_ANT_SW
I/O Expander #1 PI4IOE5V6408 (U3)
SYS_IRQ
SPI2_G17
I/O Expander #2 PI4IOE5V6408 (U7)
SYS_LED6
LCD_RST
User LED Green LED_GREEN (LED1)
1.14" IPS LCD Driver 135×240 ST7789
Capacitive Touch Controller FT6336
POWEROFF
5VOUT_EN
Step-Up Converter SGM6603 (U4)
Power LED Blue LED_BLUE (LED3)
BOOT
Power Management IC PMS150G (U5)
PWN_EN
Step-Down Converter JW5712 (U6)
SW Power Button (S3)
ESP32-C6 Buses: SPI_Flash (dedicated) for Flash only; SPI2 (G20/G21/G22) shared by LCD (CS:G17) and LoRa (CS:G23) and no simultaneous access. I2C0 (G10/G8) shared by all peripherals, Qwiic, and HAT. Reserved I2C addresses: 0x68, 0x38, 0x20, 0x21, 0x55, 0x34.
Legend:
Connector
Main Part
Internal Part
I2C/I2S
SPI
Radio/Reserved
Other



I/O Expanders (PI4IOE5V6408):

- E0 @ 0x43 (buttons & LoRa control)
- E1 @ 0x44 (power/UI: LCD reset/backlight, GROVE power, VIN detect, LED)

Legend:

- Power
- Ground
- IN Power Input
- OUT Power Output
- GPIO Digital External
- Analog External
- Main Part
- Secondary Part
- Internal Component
- Other Pins (Reset, System Control, Debugging)
- I2C — Default
- SPI — Default
- UART/USART — Default
- Other SERIAL
- Analog — Default
- PWM/Timer
- LED
- RGB LED

WARNING All GPIO pins are 3.3V logic only and are not 5V tolerant.

WARNING Handle the internal LiPo battery with care. Do not puncture, short-circuit, expose to high temperatures.

NESSO N1

ARDUINO

SKU code: TPX00227
Full Pinout - Page 1 of 6
Last update: 21 Nov, 2025

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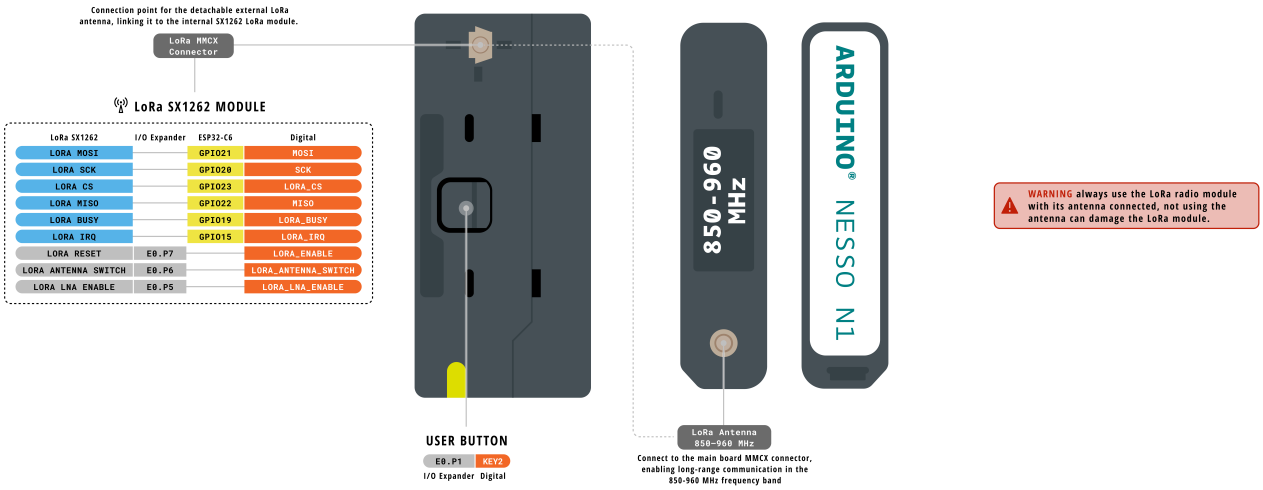

WARNING always use the LoRa radio module with its antenna connected, not using the antenna can damage the LoRa module.

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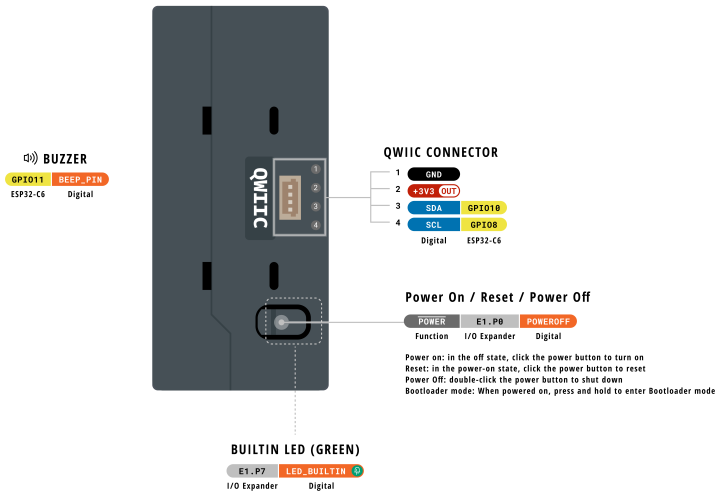

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Legend:

Power	IN Power Input	GPIO Digital External	I2C	Default	LED		
Ground	OUT Power Output	Analog External	SPI	Default	RGB LED		
		Main Part	UART/USART	Default			
		Secondary Part	Other SERIAL				
		Internal Component	Analog	Default			
		Other Pins (Reset, System Control, Debugging)	PWM/Timer				

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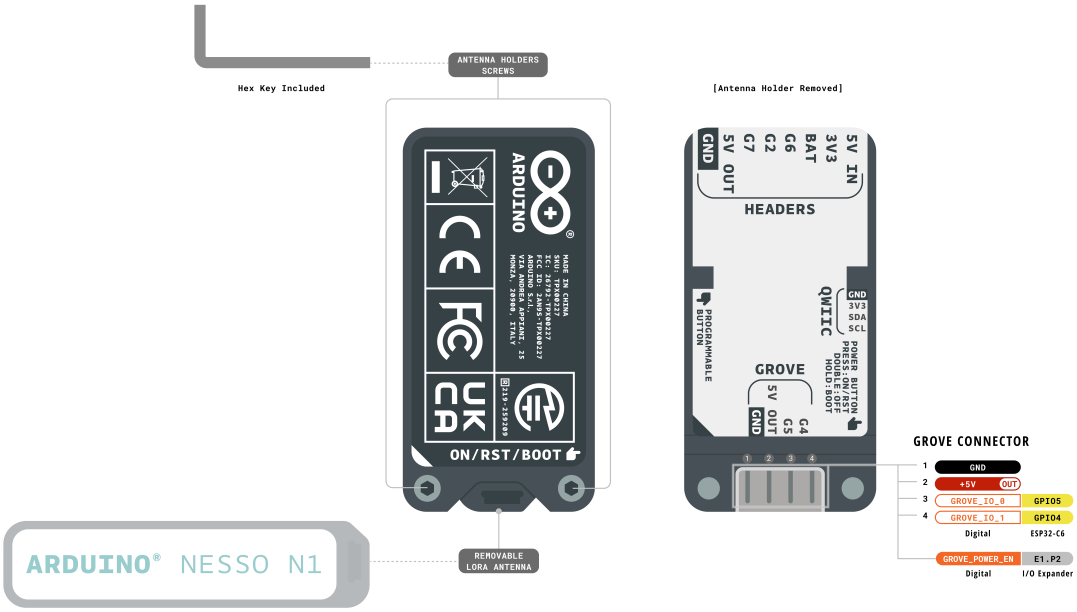

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Ground	OUT Power Output	Analog External	SPI	Default	RGB LED		
		Main Part	UART/USART	Default			
		Secondary Part	Other SERIAL				
		Internal Component	Analog	Default			
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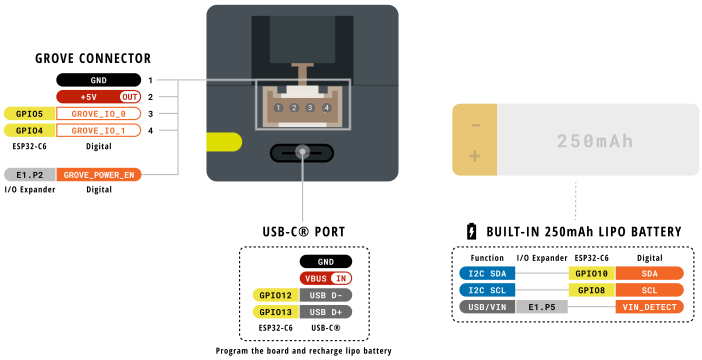

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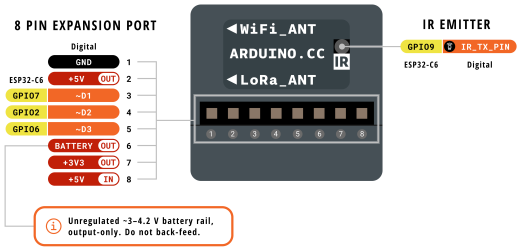

8-pin expansion port compatible with the M5StickC HAT series

Legend:

- Power
- Ground
- IN Power Input
- OUT Power Output
- GPIO Digital External
- Analog External
- Main Part
- Secondary Part
- Internal Component
- Other Pins (Reset, System Control, Debugging)
- I2C — Default
- SPI — Default
- UART/USART — Default
- Other SERIAL
- Analog — Default
- PWM/Timer
- LED
- RGB LED

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PinMap

- Arduino Nesso N1 Full Pin Map

LCD

ESP32-C6	G21	G20	G17	G16
ST7789	MOSI	SCK	CS	RS

Touch & IMU

ESP32-C6	G10	G8	G3
FT6336 (0x38)	SDA	SCL	INT
BMI270 (0x68)	SDA	SCL	INT

Power Management

ESP32-C6	G10	G8
BQ27220YZFR ()	SDA	SCL
AW32001ECSR ()	SDA	SCL

LoRa

ESP32-C6	G21	G22	G20	G23	G19	G15
SX1262	MOSI	MISO	SCK	CS	BUSY	IRQ

PI4IOE5V6408

ESP32-C6	G10	G8	ESP_RST
PI4IOE5V6408-0 (0x43)	SDA	SCL	RST
PI4IOE5V6408-1 (0x44)	SDA	SCL	RST

PI4IOE5V6408-1 (0x43)	E0.P0	E0.P1	E0.P5	E0.P6	E0.P7
KEY1	INPUT				
KEY2		INPUT			
SX1262			SX_LNA_EN	SX_ANT_SW	SX_NRST

The following steps are required for LoRa module initialization:

- Reset the LoRa module: Set SX_NRST low, hold for 100 ms, then set high.
- Enable the RF antenna switch: Set SX_ANT_SW high.
- Enable the LNA chip: Set SX_LNA_EN high.

PI4IOE5V6408-1 (0x44)	E1.P0	E1.P1	E1.P2	E1.P5	E1.P6	E1.P7
Device Power	PWR_OFF_PULSE					
LCD		LCD_RESET			LCD_BACKLIGHT	
GROVE			GROVE_POWER_EN			
USB				USB_VIN_DETECT		
LED						LED_EN

Buzzer & IR

ESP32-C6	G11	G9
Buzzer	BEEP	
IR		IR_TX

QWIIC Connector

ESP32-C6	GND	3V3	G10	G8
QWIIC	GND	3V3	SDA	SCL

HY2.0-4P

HY2.0-4P	Black	Red	Yellow	White
PORT.A	GND	5V	G5	G4

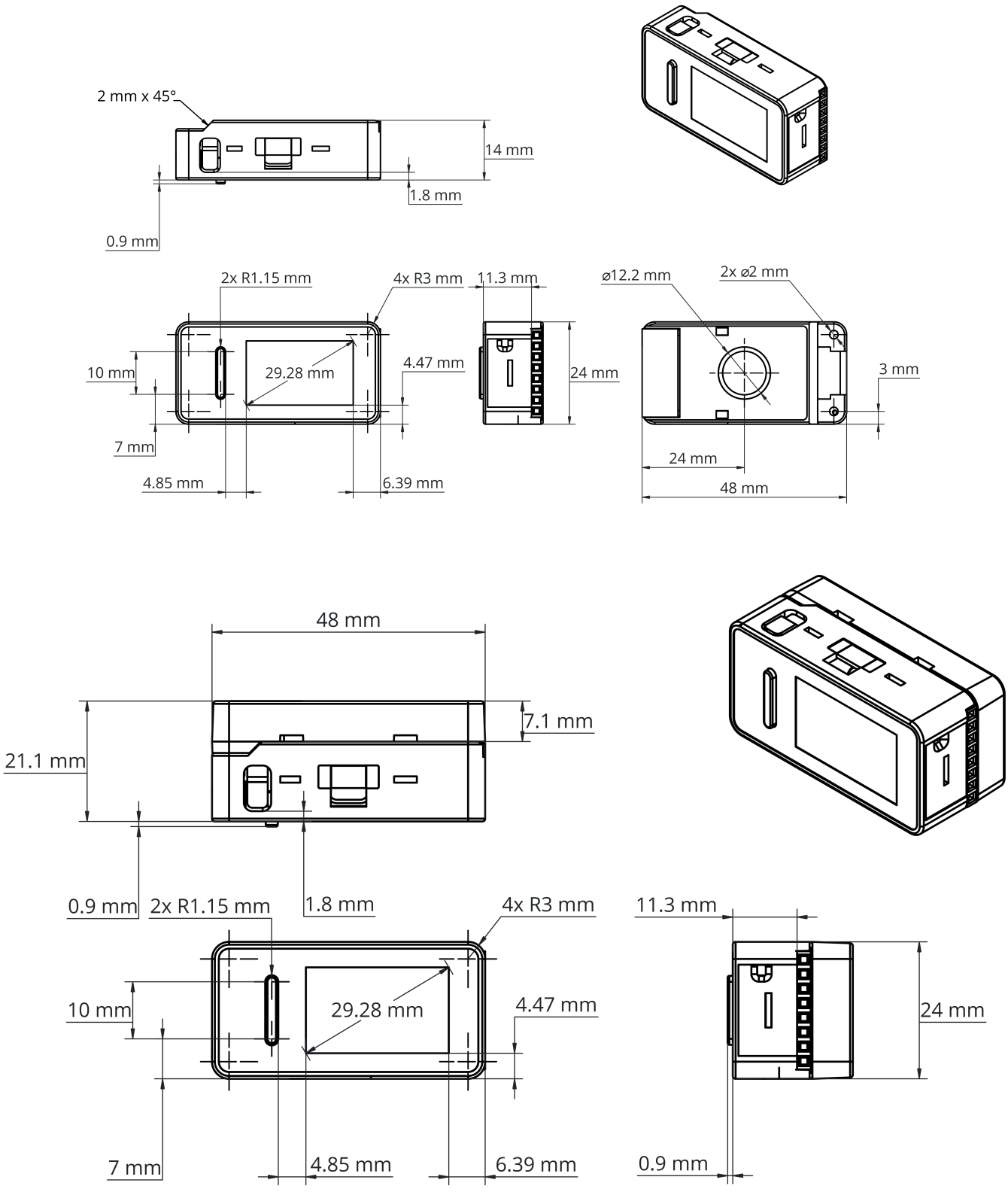
USB

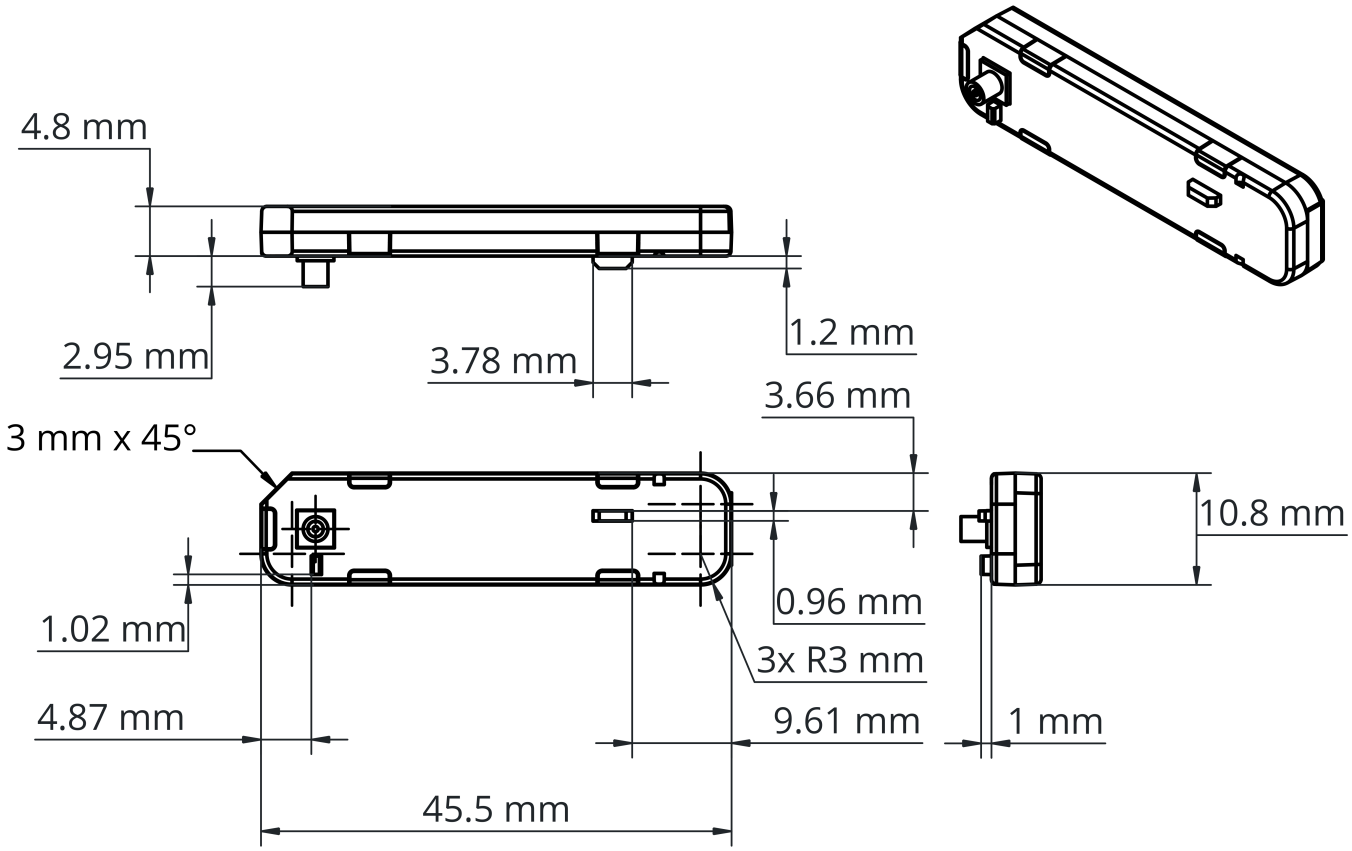
ESP32-C6	G13	G12
USB	USB_D+	USB_D-

Hat Bus

Hat Bus	Arduino Nesso N1
1	5V_IN
2	3V3
3	BAT
4	G6
5	G2
6	G7
7	5V_OUT
8	GND

Model Size


2 mm x 45°
14 mm
1.8 mm
0.9 mm
2x R1.15 mm
4x R3 mm
11.3 mm
10 mm
29.28 mm
4.47 mm
24 mm
7 mm
4.85 mm
6.39 mm
ø12.2 mm
2x ø2 mm
3 mm
24 mm
48 mm
48 mm
7.1 mm
21.1 mm
0.9 mm
2x R1.15 mm
1.8 mm
4x R3 mm
10 mm
29.28 mm
4.47 mm
7 mm
4.85 mm
6.39 mm
11.3 mm
24 mm
0.9 mm



Structure

- Arduino Nesso N1 Structure Files

Datasheets

- Arduino Nesso N1 Datasheet
- SX1262
- ESP32-C6

Softwares

Arduino

- Arduino Nesso N1 Quick Start
- Arduino Nesso N1 Official User Manual

UiFlow2

- Arduino Nesso N1 UiFlow2 Quick Start

Video